`

I Project overview

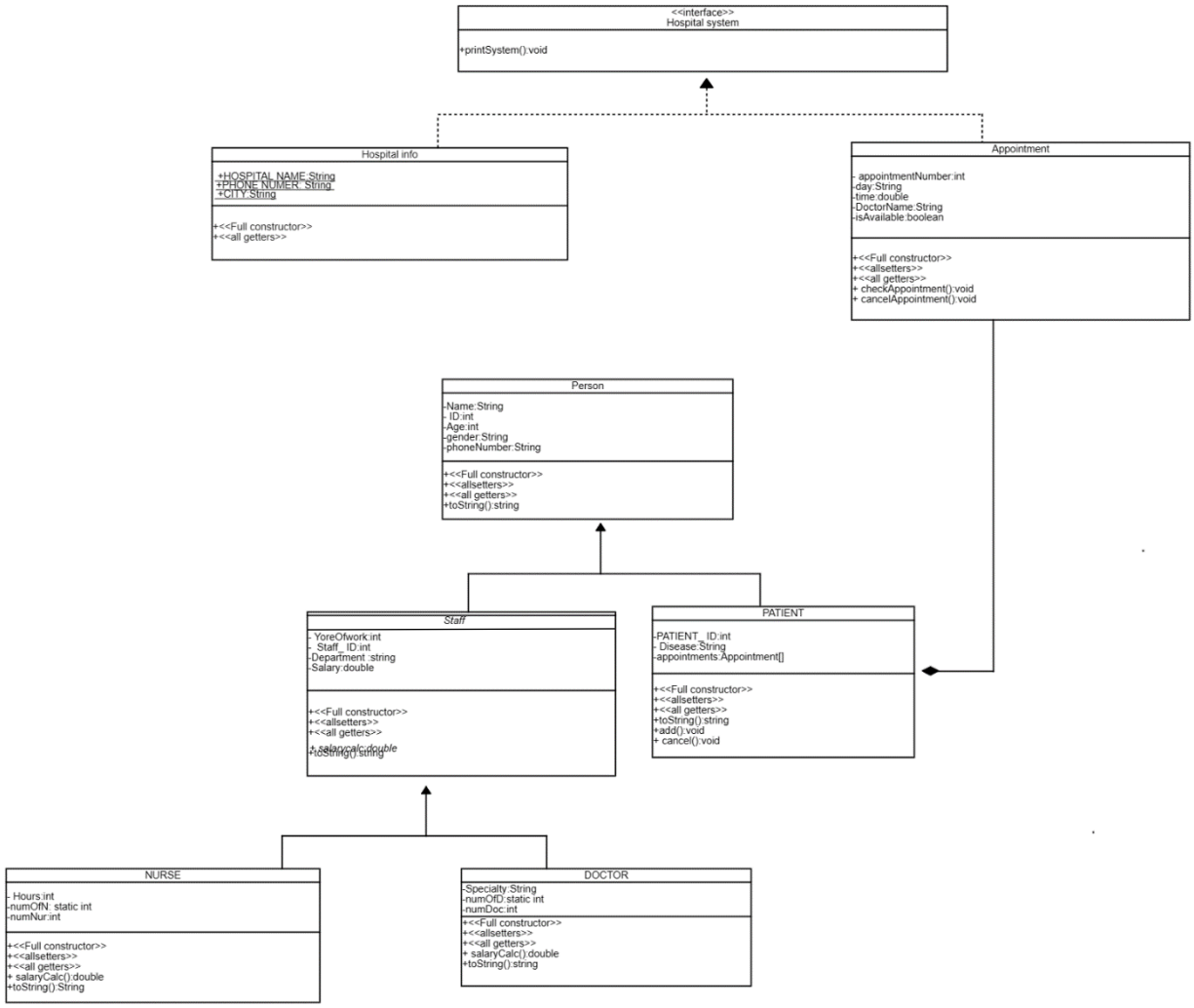
A project talking about :

Hospital System The name of our Hospital is : "Central City Hospital"

So that it includes all services that make it easier for patients

Request appointments, cancel appointments, show doctors in the system

II Project class diagram



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Hospitalsystem «interfece»> :

Will shere common method to class hospitalinfo and appointment

Hospital info :

Will implement method printSystem( ): void

This method will print all hospital information

Appointment :

This class have three method:

Will implement method printSystem() : void

This method will print all appointment information

checkAppointment(); was created to check if the appointment is available or not

CancelAppointment(); we create this method for patient if he want to cancel the appointment

Person :

Is a super class for patinent and staff method toString( ) should print all personal information

Patinent :

This class extends from class person and will override method toString()

Method add() will add number of appointments

Method cancel() will reduce the number of appointments

Staff :

Is abstract class

Is a super class for class doctor and class nurse

This class extends from class person and will override method tostring()

Method toString() will print staff information

Method salarycalc() is abstract method and the sup class should calculate the salary

Doctor :

extends from class staff and will override method toString() and salarycalc()

Method toString will print doctor information

`

Method Salarycalc() will calculate the salary if years of work is greater than or equal to three the salary will increase by 3000 .

Nurse :

extends from class staff and will override method toString() and salarycalc()

Method toString() will print nurse information

Method Salarycalc() will calculate the salary if years of work is greater than or equal to three the salary will increase by 3000 .

III Concepts covered

III.1 Interface

```
 */
package hospitalsystem;

/**
 *
 * @author Asayel
 */
public interface HospitalSystem {

    public abstract void printSystem();
}
```

III.2 Inheritance

```
 */
public final class Patient extends Person{
```

```
 */
public abstract class Staff extends Person  {
```

```
public final class Doctor extends Staff {
```

```
 */
public final class Nurse extends Staff {
```

`

III.3 Abstract classes

```
10  */
    public abstract class Staff extends Person  {
```

III.4 Polymorphism

```
System.out.println(" ****   welcome To central City Hospital  ****");
Person d1 = new Doctor("sara", 7145637, 24, "Female", "0534266767", 4, 2232, "Suger", 20000, "Brain and Nerv Surgeory");
Person d2 = new Doctor("Ahmad", 1354675, 25, "male", "0579833543", 5, 2256, "Hearing and Balance", 15000, "Hearingand Balance");
Person d3 = new Doctor("Omar", 2485990, 27, "male", "0534755543", 9, 2087, "Rays", 12000, "X-ray");
Person d4 = new Doctor("Leena", 5829431, 24, "Female", "0544237891", 3, 2288, "Eye ", 15000, "Ophthalmologist");
Person d5 = new Doctor("Asayel", 3825498, 24, "Female", "0578934557", 2, 2221, "Esotericism", 13000, "general and Esotericism ");
Person d6 = new Doctor("Shahad", 1749365, 24, "Female", "0500678424", 1, 2390, "Children", 16000, "general");
Person n1 = new Nurse("Nora", 6738291, 35, "Female", "0502333654", 5, 2234, "Emergency", 9000, 5);
Person n2 = new Nurse("Loaloah", 4352133, 20, "Female", "0552223298", 7, 2166, "Emergency", 9000, 5);
```

```
Person p = null;

p = new Patient(n, i, x, y, o, r, m, size);
```

```
System.out.println("Staff Salary :");
for (Person q : personList) {
    if (q instanceof Staff) {
        System.out.println(q);

        System.out.println("Calculate Salary: " + ((Staff) q).SalaryCalc());
        System.out.println();
    }
}
```

III.5 Composition

```
public final class Patient extends Person{

  private  int Patient_ID;
  private String Disease;
  private Appointment []appointments;
```

III.6 ArrayList (in the main)

```
public static ArrayList<Person> personList = new ArrayList<Person>();
/**
```

```
ArrayList<Appointment> AppointmentList = new ArrayList<Appointment>();
```

III.7 Array (with composition)

```
private  int Patient_ID;
private String Disease;
private Appointment []appointments;
```

III.8 exception handling

```
        int choice = 0;
        try {
            while (choice != 10) {
            }
        } catch (InputMismatchException ex) {
            scanner.next();
            System.err.println("Invalid input!");
        } catch (NullPointerException ex) {
            System.err.println("Invalid object!");
        } catch (ClassCastException ex) {
            System.err.println("Invalid downcasting!");
        }

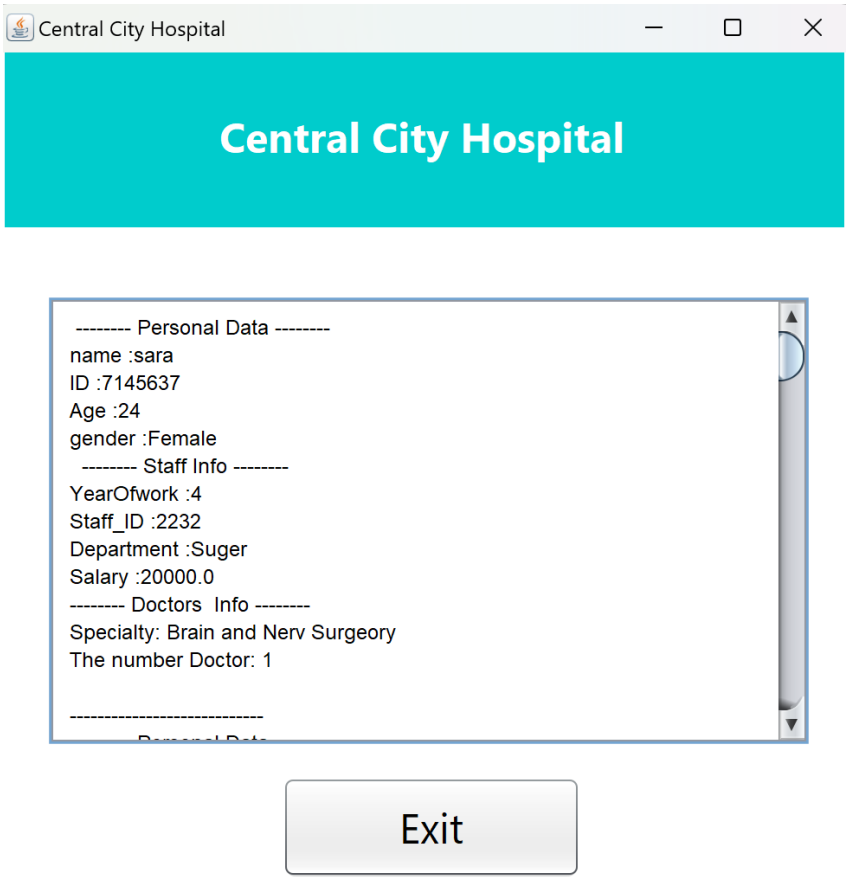
    }
```

```
    private int x ;
   public void add(Appointment s){
    try{
    appointments[x ]= s;
      x++;}
    catch(Exception m){
        System.out.println("i can't add more index !!");
    }
}
      public void cancel (int i){
    try{
            appointments[i]=appointments[x-i];
            x--;
        }catch(Exception e){
            System.out.println(" Worng index!!");
        }

    }
```

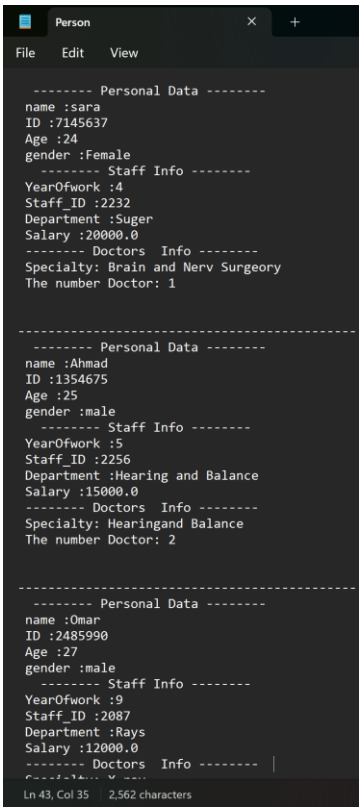
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III.9 GUI



III.10 Files

```
201
202                 case 9:
203                     int option = 0;
204                     while (option != 3) {
                            System.out.println("Choose which option do you whant:\n"
206                                 + "1- WriteForm File\n"
207                                 + "2- Read From File\n"
208                                 + "3- Exit From File ");
209                         option = scanner.nextInt();
                            switch (option) {
211
212                             case 1:
213                                 Write e = new Write();
214                                 e.openTextFile("Person.txt");
215                                 for (Person j : personList) {
216                                     e.writeToFile(j);
217                                 }
218
219                                 e.closeFile();
220                                 System.out.println("Data has been saved to Person.txt");
221                                 break;
222
223                             case 2:
224                                 Read t = new Read();
225                                 t.openTextFile("Person.txt");
226                                 t.read_File();
227                                 t.closeFile();
228
229                                 break;
230                             case 3:
231                                 System.out.println("Thank you");
232                                 System.out.println();
233                                 break;
```



`

IV Main Class

We creat ArrayList form Class Person And we add 8 object to the arrayList 6 from Doctor and 2 from Nurse

and we creat ArrayList from Class Appointment we need it for number of appointments After that we print menu for user to enter his options each case have a Different properties.

Case 1 :

Print all hospital information

Case 2:

This case makes the patient fill in his data himself as well. If the entry is less than zero, that is, negative, he will send a message "Invalid ID" as well, and then if patient finish it will output the message "Patient has been created".

Case 3:

The user should add the day than time and The user will add the number of appointment after that we print a list of doctors the user choose one doctor than the user enter true if he finish the appointment information or false if he want to exit from add the appointment

Than we print statement of all information about appointment

Case 4:

If The user choose this case will print is Available if he write true in add appointment else will print is available for book

Case 5 :

This case appears to be part of a program that cancels appointments. If the "a" object is of type

"appointment", this appointment will be canceled using the "cancelAppointment" method.

case 6 :

will print all information about class staff

Case 7:

This case calculates the salary. If the "" object is of type "Staff", the salary will be calculated for the staff. After calculating the salary, the new salary for the staff will be printed.

case 8 :

will show you the output of gui

`

case 9 :

will get you 3 options the option one will write file option two read file option three will close file

case 10:

will close the code



```
import java.io.File;
import java.io.FileNotFoundException;
import java.util.Scanner;
import java.util.ArrayList;
import java.util.FormatterClosedException;
import java.util.InputMismatchException;
import java.util.NoSuchElementException;

public class Main {

    Scanner scanner = new Scanner(System.in);
    public static ArrayList<Person> personList = new ArrayList<Person>();
    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) throws FileNotFoundException {

        Scanner scanner = new Scanner(System.in);
        Person p = null;

        Appointment ap = null;
        int size = 0;

        ArrayList<Appointment> AppointmentList = new ArrayList<Appointment>();

        System.out.println(" ****   welcome To central City Hospital  ****");
        Person d1 = new Doctor("Sara", 7145637, 24, "Female", "0534266767", 4, 2232, "Suger", 20000, "Brain and Nerv Surgeory");
        Person d2 = new Doctor("Ahmad", 1354675, 25, "male", "0579833543", 5, 2256, "Hearing and Balance", 15000, "Hearingand Balance");
        Person d3 = new Doctor("Omar", 2485990, 27, "male", "0534755543", 9, 2087, "Rays", 12000, "X-ray");
        Person d4 = new Doctor("Leena", 5029431, 24, "Female", "054423709l", 3, 2200, "Eye ", 13000, "Opthalmologist");
        Person d5 = new Doctor("Asayel", 3825498, 24, "Female", "0578934557", 2, 2221, "Esotericism", 13000, "general and Esotericism ");
        Person d6 = new Doctor("Shahad", 1749365, 24, "Female", "0500678424", 1, 2390, "Children", 16000, "general");
        Person n1 = new Nurse("Nora", 6730291, 30, "Female", "0502333604", 8, 2234, "Emergency", 9000, 5);
        Person n2 = new Nurse("Loaloah", 4352133, 20, "Female", "0552223298", 7, 2166, "Emergency", 9000, 5);

        personList.add(d1);
        personList.add(d2);
        personList.add(d3);
```

```
        personList.add(d4);
        personList.add(d5);
        personList.add(d6);
        System.out.println();
        personList.add(n1);
        personList.add(n2);

        System.out.println("please choose one of the following options : ");
        int choice = 0;
        try {
            while (choice != 10) {
                System.out.println("""
                    1- Display Hospital Info
                    2- Fill Patient Information
                    3- Add Appointment
                    4- Chack Appointment
                    5- Cancel Appointment
                    6- Show staff info
                    7- For Salary this option for Staff only !!
                    8- GUI
                    9- File
                    10- Exit """);
                choice = scanner.nextInt();
                switch (choice) {
                    case 1:
                        System.out.println("_____ hopsital info _______");
                        HospitalInfo k = new HospitalInfo();
                        k.printSystem();
                        System.out.println();
                        break;
                    case 2:

                        System.out.println(" __enter information of Patient __");
                        System.out.println("Enter your name :");
                        String n = scanner.next();

                        System.out.println("Enter your ID :");
                        int i = scanner.nextInt();
                        if (i <= 0) {
```

```
                    System.out.println("invalid ID");
                }
                System.out.println("Entre your Age : ");
                int x = scanner.nextInt();

                System.out.println("Enter your gender : ");
                String y = scanner.next();
                System.out.println("Enter your phone Number : ");
                String o = scanner.next();
                System.out.println("Enter patient  file ID :");
                int r = scanner.nextInt();
                System.out.println("Enter your Disease :");
                String m = scanner.next();
                System.out.println("how many  appointment you whant ? ");
                size = scanner.nextInt();
                System.out.println();
                p = new Patient(n, i, x, y, o, r, m, size);

                System.out.println("Patient has been created ");
                personList.add(p);

                System.out.println(p.toString());
                System.out.println("---------------------------------");
                System.out.println();
                break;

            case 3:
                System.out.println("___ add appointment ___");
                for (int j = 1; j <= size; j++) {

                    System.out.println("Appointment Number " + j);
                    System.out.println("enter the day ");
                    String s = scanner.next();
                    System.out.println("enter time ");
                    double t = scanner.nextDouble();

                    System.out.println("enter appointment Number");
                    int mm = scanner.nextInt();
                    System.out.println(
```

```
                    System.out.println(
                            "choose the Doctors name :\n 1- Dr.Sara \n 2- Dr.Ahmad \n 3-Dr.Omar \n 4-Dr.Leena \n 5-Dr.Asayel \n 6-Dr.Shahd ");
                    System.out.println("enter  your doctor Name");
                    String d = scanner.next();
                    System.out.println("enter true if you finsh or false to cancel");
                    boolean uo = scanner.nextBoolean();
                    if (uo == true) {

                        System.out.println("your appointments has been  creted");
                    } else {
                        System.out.println("invalid appointment !");
                    }
                    as = new Appointment(s, t, mm, d, uo);
                    AppointmentList.add(as);

                }
                as.printSystem();

                System.out.println();
                break;
            case 4:
                System.out.println(" ___ chack Appointment ___");
                for (Appointment q : AppointmentList) {
                    if (q instanceof Appointment) {
                        q.CheekApoointment();
                    }
                }

                System.out.println();
                break;
            case 5:
                System.out.println("___ cancel appointment ___ ");
                for (Appointment q : AppointmentList) {
                    if (q instanceof Appointment) {
                        q.cancelAppointment();
                    }
                }
                System.out.println();
                break;
```

`

```
                case 6:
                    System.out.println();
                    for (Person w : personList) {
                        if (w instanceof Staff) {
                            System.out.println(w.toString());


                            System.out.println("_____________________________________________________________________________");
                        }
                    }


                    System.out.println();
                    break;

                case 7:
                    System.out.println("Staff Salary :");
                    for (Person q : personList) {
                        if (q instanceof Staff) {
                            System.out.println(q);

                            System.out.println("Calculate Salary: " + ((Staff) q).SalaryCalc());
                            System.out.println();
                        }
                    }

                    System.out.println();
                    break;

                case 8:
                    // GUI

                    GUI.main(args);
                    System.out.println();
                    break;

                case 9:
                    int option = 0;
                    while (option != 3) {
```

```
                        System.out.println("Choose which option do you whant:\n"
                                + "1- WriteForm File\n"
                                + "2- Read From File\n"
                                + "3- Exit From File ");
                        option = scanner.nextInt();
                        switch (option) {

                            case 1:
                                Write e = new Write();
                                e.openTextFile("Person.txt");
                                for (Person j : personList) {
                                    e.writeToFile(j);
                                }

                                e.closeFile();
                                System.out.println("Data has been saved to Petson.txt");
                                break;

                            case 2:
                                Read t = new Read();
                                t.openTextFile("Person.txt");
                                t.read_File();
                                t.closeFile();

                                break;
                            case 3:
                                System.out.println("Thank you");
                                System.out.println();
                                break;

                            default:
                                System.out.println("Invalid Option!");
                                System.out.println();
                        }
                    }
                    // file

                    System.out.println();
                    break;
```

`

```
                    case 10:

                        System.out.println("thank you for using our system , we hope you enjoyed it (: ");
                        System.out.println();
                        break;

                    default:
                        System.out.println(" invalid choice >_< ");
                        System.out.println(" please choose a number from the menu  : ");
                        System.out.println();
                        break;
                }
            }
        } catch (InputMismatchException ex) {
            scanner.next();
            System.err.println("Invalid input!");
        } catch (NullPointerException ex) {
            System.err.println("Invalid object!");
        } catch (ClassCastException ex) {
            System.err.println("Invalid downcasting!");
        }

    }

}  //main class
```

The OutPut:

```
run:
 ****   welcome To central City Hospital  ****

please choose one of the following options :
1- Display Hospital Info
2- Fill Patient Information
3- Add Appointment
4- Chack Appointment
5- Cancel Appointment
6- Show staff info
7- For Salary this option for Staff only !!
8- GUI
9- File
10- Exit
```

`

```
1
_______ hopsital info ________
hospital Name :  Central City Hospital
city : Riyadh
Hospital Phone : 0116498


2
 __enter information of Patient __
Enter your name :
Asayel
Enter your ID :
1128887989
Entre your Age :
19
Enter your gender :
Female
Enter your phone Number :
05548321
Enter patient  file ID :
8844
Enter your Disease :
EarPain
how many  appointment you whant ?
1

Patient has been created
 -------- Personal Data --------
 name :Asayel
 ID :1128887989
 Age :19
 gender :Female
 Patient_ID :8844
Disease :EarPain
--------------------------------
```

`

```
3
___ add appointment ___
Appointment Number 1
enter the day
Sunday
enter time
2.5
enter appointment Number
1
choose the Doctors name :
 1- Dr.Sara
 2- Dr.Ahmad
 3-Dr.Omar
 4-Dr.Leena
 5-Dr.Asayel
 6-Dr.Shahd
enter  your doctor Name
sara
enter true if you finsh or false to cancel
true
your appointments has been  creted
----- Appointment Information -----
The Day of Appointment is: Sunday
The Time is : 2.5
Appointment Number :1
Doctor Name :sara
```

```
--  ----
4
 ___ chack Appointment ___
The Appointment is Available
```

```
5
___ cancel appointment ___
The Appointment has been canceled and is availabl now
```

```
`
6

 -------- Personal Data --------
name :sara
ID :7145637
Age :24
gender :Female
  -------- Staff Info --------
YearOfwork :4
Staff_ID :2232
Department :Suger
Salary :20000.0
-------- Doctors  Info --------
Specialty: Brain and Nerv Surgeory
The number Doctor: 1


 -------- Personal Data --------
name :Ahmad
ID :1354675
Age :25
gender :male
  -------- Staff Info --------
YearOfwork :5
Staff_ID :2256
Department :Hearing and Balance
Salary :15000.0
-------- Doctors  Info --------
Specialty: Hearingand Balance
The number Doctor: 2


 -------- Personal Data --------
name :Omar
ID :2485990
Age :27
gender :male
  -------- Staff Info --------
YearOfwork :9
Staff_ID :2087
Department :Rays
Salary :12000.0
-------- Doctors  Info --------
Specialty: X-ray
The number Doctor: 3


 -------- Personal Data --------
name :Leena
ID :5829431
Age :24
gender :Female
  -------- Staff Info --------
YearOfwork :3
Staff_ID :2288
Department :Eye
Salary :15000.0
-------- Doctors  Info --------
Specialty: Ophthalmologist
The number Doctor: 4


 -------- Personal Data --------
name :Asayel
ID :3825498
Age :24
gender :Female
  -------- Staff Info --------
YearOfwork :2
Staff_ID :2221
Department :Esotericism
Salary :13000.0
```

`

```
7
Staff Salary :
  -------- Personal Data --------
 name :sara
 ID :7145637
 Age :24
 gender :Female
   -------- Staff Info --------
 YearOfwork :4
 Staff_ID :2232
 Department :Suger
 Salary :20000.0
 -------- Doctors  Info --------
 Specialty: Brain and Nerv Surgeory
 The number Doctor: 1

Calculate Salary: 23000.0

  -------- Personal Data --------
 name :Ahmad
 ID :1354675
 Age :25
 gender :male
   -------- Staff Info --------
 YearOfwork :5
 Staff_ID :2256
 Department :Hearing and Balance
 Salary :15000.0
 -------- Doctors  Info --------
 Specialty: Hearingand Balance
 The number Doctor: 2

Calculate Salary: 18000.0

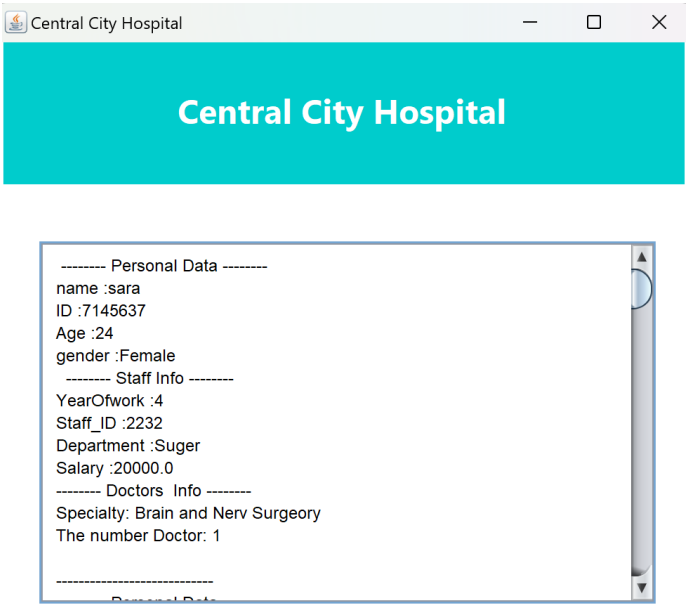
  -------- Personal Data --------
 name :Omar
 ID :2485990
 Age :27
 gender :male
   -------- Staff Info --------
 YearOfwork :9
 Staff_ID :2087
 Department :Rays
 Salary :12000.0
 -------- Doctors  Info --------
 Specialty: X-ray
 The number Doctor: 3

Calculate Salary: 15000.0

  -------- Personal Data --------
 name :Leena
 ID :5829431
 Age :24
 gender :Female
   -------- Staff Info --------
 YearOfwork :3
 Staff_ID :2288
 Department :Eye
 Salary :15000.0
 -------- Doctors  Info --------
 Specialty: Ophthalmologist
 The number Doctor: 4

Calculate Salary: 18000.0

  -------- Personal Data --------
 name :Asayel
 ID :3825498
 Age :24
 gender :Female
   -------- Staff Info --------
```




```
9
Choose which option do you whant:
1- WriteForm File
2- Read From File
3- Exit From File
1
Data has been saved to Petson.txt
```

```
-----------------------------------------------
Choose which option do you whant:
1- WriteForm File
2- Read From File
3- Exit From File
3
Thank you
```

```
Data has been saved to Petson.txt
Choose which option do you whant:
1- WriteForm File
2- Read From File
3- Exit From File
2
  -------- Personal Data --------
 name :sara
 ID :7145637
 Age :24
 gender :Female
   -------- Staff Info --------
 YearOfwork :4
 Staff_ID :2232
 Department :Suger
 Salary :20000.0
 -------- Doctors  Info --------
 Specialty: Brain and Nerv Surgeory
 The number Doctor: 1


-----------------------------------------------
  -------- Personal Data --------
 name :Ahmad
 ID :1354675
 Age :25
 gender :male
   -------- Staff Info --------
 YearOfwork :5
 Staff_ID :2256
 Department :Hearing and Balance
 Salary :15000.0
 -------- Doctors  Info --------
 Specialty: Hearingand Balance
 The number Doctor: 2


-----------------------------------------------
  -------- Personal Data --------
 name :Omar
 ID :2485990
 Age :27
 gender :male
   -------- Staff Info --------
 YearOfwork :9
 Staff_ID :2087
 Department :Rays
 Salary :12000.0
 -------- Doctors  Info --------
 Specialty: X-ray
 The number Doctor: 3
```

```
1- Display Hospital Info
2- Fill Patient Information
3- Add Appointment
4- Chack Appointment
5- Cancel Appointment
6- Show staff info
7- For Salary this option for Staff only !!
8- GUI
9- File
10- Exit
10
thank you for using our system , we hope you enjoyed it (:
```

V Work distribution

		لينا عبدالاله السرحان		
Interface		----------	--	-
Inheritance	--	Class staff extends from person And class Doctor extends from staff		
Abstract classes	-- -	Abstract class staff	-	
Polymorphism	C A	Create object And salaryCalc	C re A nd	lc
Composition	C ent	----------	-- ---	
ArrayList (in the main)	on	Array List Person	A ay L st	
Array (with composition)	--- --	----------	C An	
exception handling	All ception c as	In class Appointment	-------	
Files	W s and Read files	Witer Fails and Read files	----------	----------